

MACRO-I

```
#include <stdio.h>

#include <conio.h>

#include <string.h>


struct mnt1 {

    char name[15];

    int npp, nkp, nev, mdtp, kpdp, sstp;

} mnt;


char pntab[5][15]; // PNTAB (Parameter Name Table)

char mdt[50][50]; // MDT (Macro Definition Table)


void main() {

    FILE *f1;

    char ch[80], *p;

    int i, pntbptr = 0, mdtptr = 1;


    // Initialize MNT values

    mnt.npp = mnt.nkp = mnt.nev = 0;

    mnt.mdtp = 1;

    mnt.kpdp = 0;

    mnt.sstp = 0;


    clrscr();

    f1 = fopen("MAC.txt", "r");

    fgets(ch, 80, f1);

    p = strtok(ch, " ");

    if (p[strlen(p) - 1] == '\n')

        p[strlen(p) - 1] = '\0';
```

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if (strcmp(p, "MACRO") == 0) {

    // Read Macro Name

    fgets(ch, 80, f1);

    p = strtok(ch, " ,\n");

    strcpy(mnt.name, p);


    // Read Macro Parameters

    p = strtok(NULL, " ,\n");

    while (p) {

        strcpy(pntab[pntbptr], p);

        pntbptr++;

        mnt.npp++;

        p = strtok(NULL, " ,\n");

    }


    // Process Macro Body

    while (fgets(ch, 80, f1)) {

        if (strcmp(ch, "MEND\n") == 0) {

            strcpy(mdt[mdtptr], "MEND");

            mdtptr++;

            break;

        }


        p = strtok(ch, " ,\n");

        strcpy(mdt[mdtptr], "");

        while (p) {

            int found = 0;

            for (i = 0; i < pntbptr; i++) {

                if (strcmp(p, pntab[i]) == 0) {

                    char temp[10];

                    sprintf(temp, "(P,%d)", i + 1);

```

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        strcat(mdt[mdtptr], temp);

        found = 1;

        break;
    }
}

if (!found) {
    strcat(mdt[mdtptr], p);
}

strcat(mdt[mdtptr], " ");

p = strtok(NULL, ",\n");
}

mdtptr++;
}

```

// Print PNTAB

```

printf("\t\tPNTAB\n");
printf("-----\n");
for (i = 0; i < pntbptr; i++) {
    printf("%d | %s\n", i + 1, pntab[i]);
}
printf("-----\n\n");

```

// Print MNT

```

printf("\t\t MNT\n");
printf("-----\n");
printf(" Name   #pp #kp #ev MDTP KPDTP SSTP\n");
printf("%s   %d  %d  %d  %d  %d  %d",
        mnt.name, mnt.npp, mnt.nkp, mnt.nev, mnt.mdtp, mnt.kpdtp, mnt.sstp);
printf("\n-----\n");

```

// Print MDT

```
printf("\n\t\tMDT\n");

printf("-----\n");

for (i = 1; i < mdtptr; i++) {
    printf("%d | %s\n", i, mdt[i]);
}

printf("-----\n");
} else {
    printf("Invalid source...\n");
}

fclose(f1);

getch();
}
```